

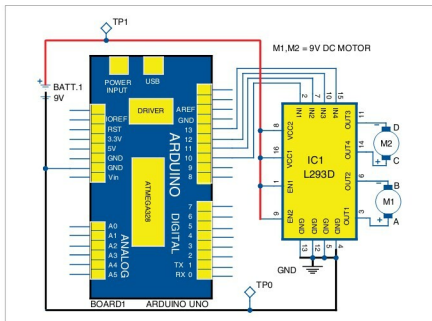


Fig. 3: Circuit diagram for controlling the robotic car through MATLAB based GUI

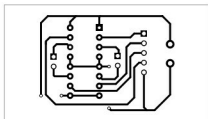


Fig. 4: Actual-size PCB pattern for the circuit to control the robotic car

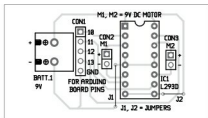


Fig. 5: Component layout of the PCB

lation directory of the IDE).

After pasting pde folder in correct location, open Arduino IDE. If you have pasted the folder in correct location, you would find pde in File→Examples→pde. Open the code by File→Examples→pde→adidos.

3. Connect Arduino Uno board to your PC. From Device Manager, note the COM port at which Arduino Uno board is installed. From Tools menu in Arduino IDE, select the board as

TABLE II TEST POINTS

Test point	Details
TP0	0V (GND)
TP1	9V

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno
IC1	- L293D motor driver
Miscellaneous:	
CON1	- 5-pin beng strip connector for Arduino pins
CON2, CON3	- 2-pin connector
M1, M2	- 9V DC motor
BATT.1	- 9V battery
	- 2-pin terminal connector

Arduino Uno and select the COM Port number noted earlier. Upload adidos code to Arduino Uno board by pressing Upload button in the IDE.

4. Copy the entire contents of the extracted ArduinoIO folder to a folder in My Documents (For Windows PC).

5. Our application program has been developed in R2014a version of MATLAB. After installing this version of MATLAB in your PC, open install_arduino.m file present in the directory where you have copied the contents of ArduinoIO folder. Now,

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

run install_arduino.m file. This code will correctly install and save the path of Arduino support package.

6. Open the source code file for this project titled car.m. Edit the COM port [in the line a = arduino('COM9')] with the port number in your PC where Arduino Uno board has been installed. After running the file, MATLAB will try to communicate with the board.

After successful communication is established, you can control the robot by pressing the appropriate pushbutton in the GUI.

Constructing and testing

An actual-size, single-side PCB for the circuit controlling the robotic car using MATLAB based GUI using Arduino is shown in Fig. 4 and its component layout in Fig. 5. Suitable connectors are provided on the PCB for input (CON1) from Arduino and output (CON2 and CON3) for motors.

Assemble the robotic car by connecting two motors, two wheels (rear) and a castor wheel (front). Connect the circuit as per the circuit diagram shown in Fig. 3. Connect the battery with correct polarity; otherwise you might end up damaging Arduino board and L293D. For troubleshooting, check the voltages at various test points given in Table II. **EFY**

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